Class Exercise.

Practice Problem 1

Consider an incident wave moving in the positive x direction on a string. When the wave reaches the boundary of that string, it continues to travel on another string which has twice the mass density of the 1st string. Determine the values for the reflection & transmission coefficients.

[Both strings are subjected to the same tension.]

(1)

Solution

Reflection coefficient: $r = \left[\frac{Z_1 - Z_2}{Z_1 + Z_2}\right]$

$Z = \sqrt{\tau\mu}$

$\tau_1, \mu_1 = \mu$ $\quad$ τ_2 $\quad$ $\mu_2 = 2\mu$

$$r = \frac{\sqrt{\mu_1\tau_1} - \sqrt{\mu_2\tau_2}}{\sqrt{\mu_1\tau_1} + \sqrt{\mu_2\tau_2}}$$

$\tau_1 = \tau_2 = \tau$

$$r = \frac{\sqrt{\tau\mu}\,[1 - \sqrt{2}\,]}{\sqrt{\tau\mu}\,[1 + \sqrt{2}\,]} = -0.172$$

$$t = \frac{2Z_1}{Z_1 + Z_2} = \frac{2\sqrt{\tau\mu}}{\sqrt{\tau\mu}\,[1 + \sqrt{2}]}$$

$$t = 0.828$$

Practice Problem 2

An acoustical foam has a density of 30 kg/m^3. Assuming air density to be 1-2 kg/m^3 with the bulk modulus of $B_a = 1.42 \times 10^5 N/m^2$, determine the bulk modulus of the foam such that it reflects none of sound normally incident on it.

Solution:

Impedance for fluids or foam : $Z = \sqrt{\rho B}$

For perfect absorption : $r = 0$

$$\therefore r = \frac{Z_1 - Z_2}{Z_1 + Z_2} = 0 \rightarrow Z_1 = Z_2$$

Since $Z_1 = Z_2$

$$\sqrt{B_f \rho_f} = \sqrt{B_a \rho_a}$$

$$\therefore B_f = \frac{B_a \rho_a}{\rho_f}$$

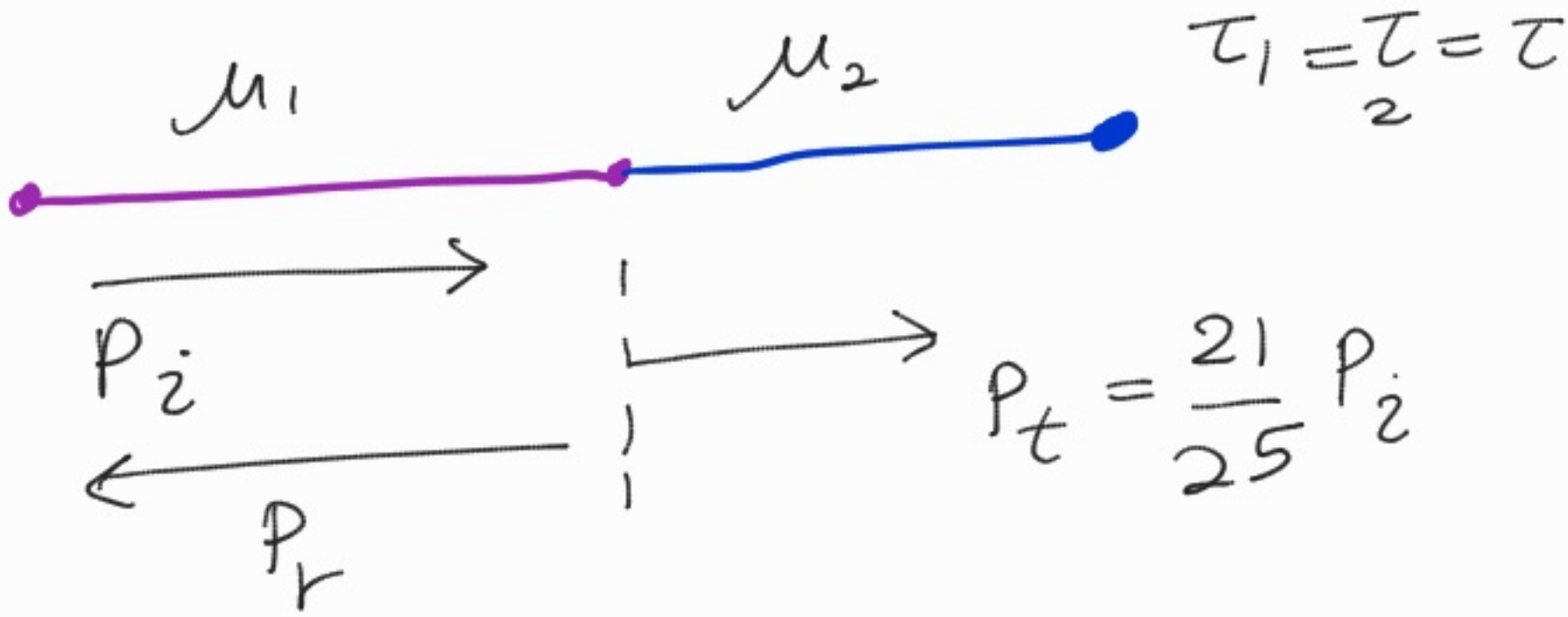
$$= \frac{(1.42 \times 10^5)(1.2)}{30} Pa$$

$$= \underline{5680 Pa}$$

Practice Problem 3 (a)

Consider an incident wave moving in the positive x direction on a string. When the wave reaches the boundary of that string, it continues to travel on another less dense string which has a different mass density compared to the 1st string. Both strings are subjected to the same tension. The incident wave is sinusoidal on the 1st string carrying power, P. When it reaches the 2nd string, its power becomes $\frac{21}{25}$ P. What is the ratio between the mass densities of both strings?

Solution:



Determine Reflected Power:

$$P_r = P_i - P_t = \left(1 - \frac{21}{25}\right) P_i = \frac{4}{25} P_i$$

Reflection Coefficient: r

$$P_r = r^2 P_i \rightarrow r^2 = \left[\frac{Z_1 - Z_2}{Z_1 + Z_2}\right]^2$$

$$r^2 = \frac{4}{25} \rightarrow r = \pm \frac{2}{5} = \frac{Z_1 - Z_2}{Z_1 + Z_2}$$

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Solution (Con't)

$$r = \frac{2}{5} = \pm \frac{Z_1 - Z_2}{Z_1 + Z_2} \quad \Big| \quad Z = \sqrt{\mu \tau}$$

Since the second string is less dense:

$\mu_1 > \mu_2 \rightarrow Z_1 > Z_2$

when $\tau_1 = \tau_2 = \tau$. Thus we take the positive solution:

$$\frac{2}{5}(Z_1 + Z_2) = Z_1 - Z_2$$

$$-\frac{3}{5} Z_1 = -\frac{7}{5} Z_2 \rightarrow Z_1 = \frac{7}{3} Z_2$$

$$\sqrt{\mu_1 \tau} = \frac{7}{3} \sqrt{\mu_2 \tau}$$

$$\sqrt{\frac{\mu_1}{\mu_2}} = \frac{7}{3} \rightarrow \frac{\mu_1}{\mu_2} = \frac{49}{9}$$

Practice Problem 3 (b)

Determine the mass density ratios if the second string is denser than the first.

Solution.

$$r = \frac{2}{5} = \pm \frac{Z_1 - Z_2}{Z_1 + Z_2}$$

$\Rightarrow$ Take the negative solution.

$$2(Z_1 + Z_2) = 5(Z_2 - Z_1)$$

$$7Z_1 = 3Z_2$$

$$7\sqrt{\mu_1 \epsilon} = 3\sqrt{\mu_2 \epsilon}$$

$$\frac{\mu_1}{\mu_2} = \left(\frac{3}{7}\right)^2 = \frac{9}{49}$$